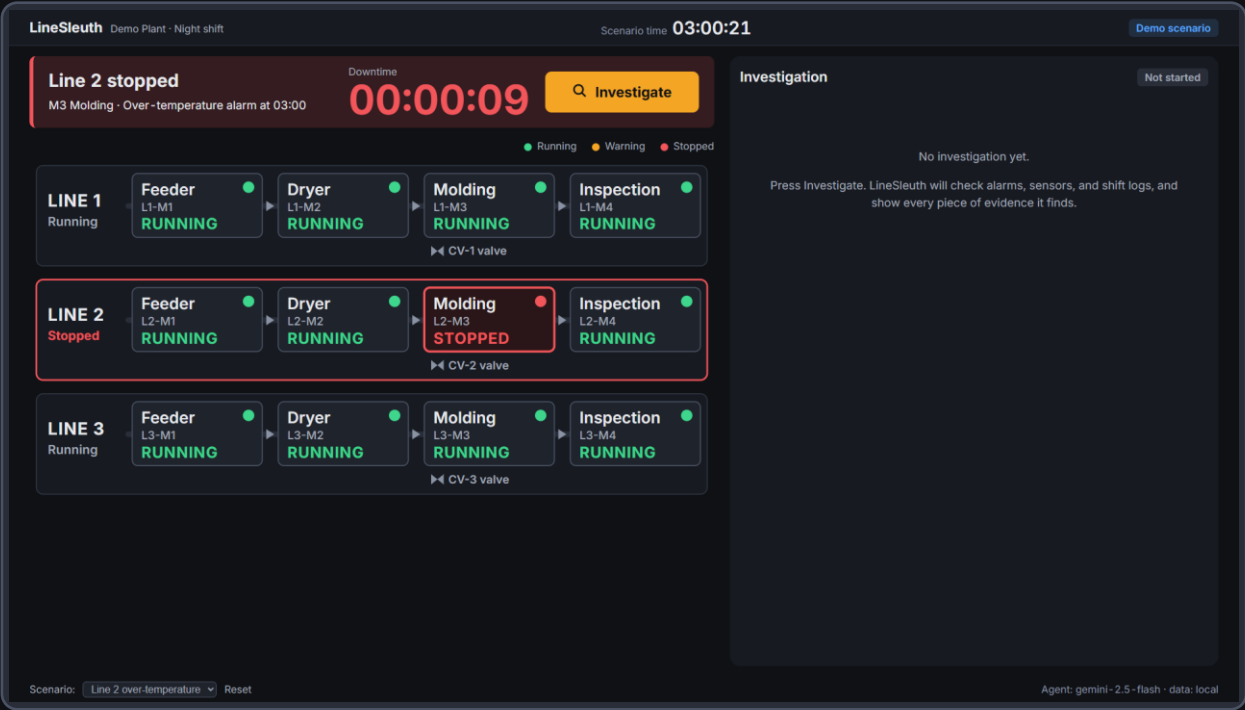


LineSleuth

Root-cause investigations for small factories, with evidence you can check.



THE PROBLEM

03:00

Line 2 stops.

- Every hour of downtime costs **X***
- Finding the cause: **~40 min*** of Excel and PLC logs
- The night supervisor is alone

Line2_temps.xlsx

PLC_export_0300.csv

shift_log.xlsx

alarms_L2.csv

** pending validation: plant-manager interviews not yet completed.*

Who we serve

Night-shift supervisor

Contract manufacturer, 50–300 staff*

- No MES; data lives in Excel sheets and PLC CSV exports
- At night, decides alone; engineers are off site
- Every stoppage is a manual log hunt

“China plus one” is moving production south

Vietnam

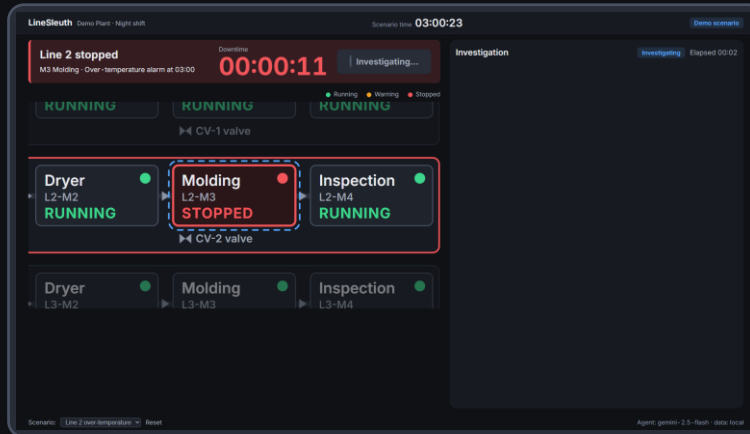
Thailand

Malaysia

Taiwan

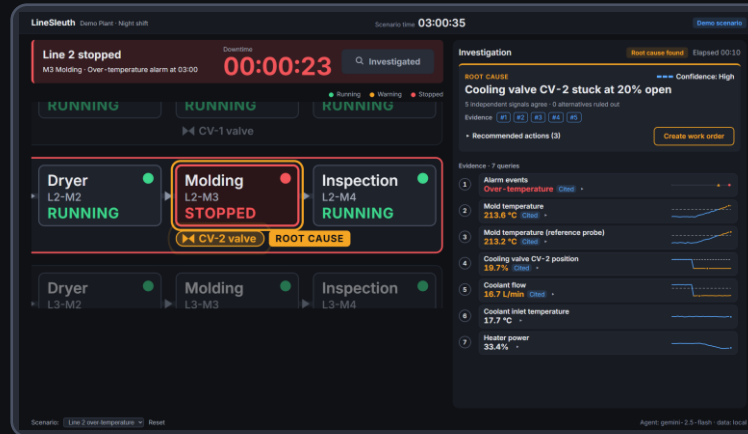
Small plants the big industrial platforms don't reach.

One button: evidence, root cause, work order



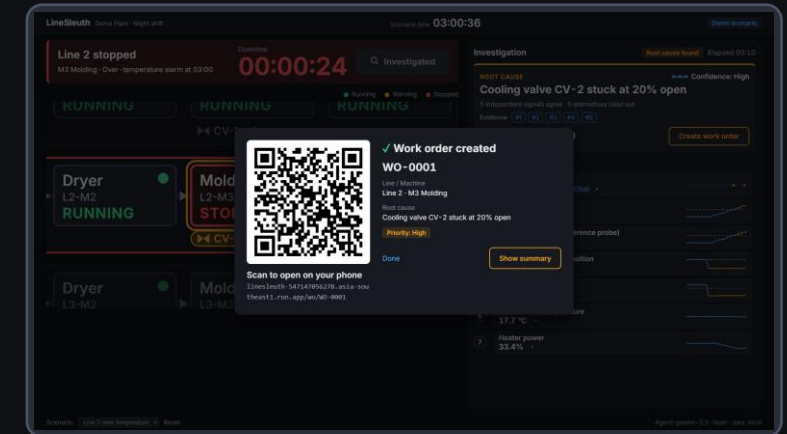
1 Press Investigate

No chat box, no prompt



2 Evidence → root cause

Every card traces back to source rows

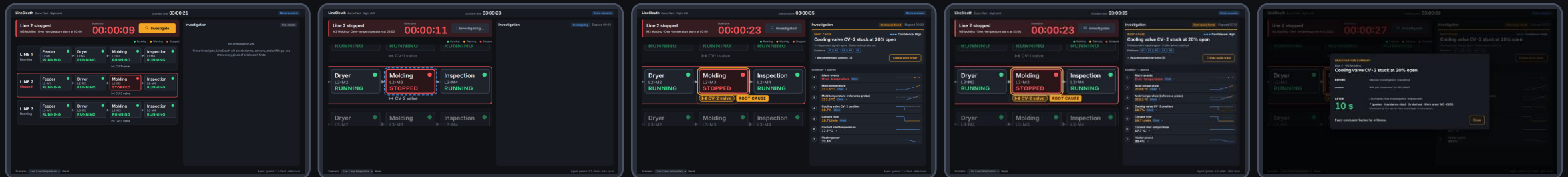


3 Work order by QR

On the technician's phone

Gemini picks from 5 fixed, tested queries — it never writes SQL. **Confidence is scored by the server.**

Live: Line 2, 03:00



1

03:00 — Line 2 stops.

2

One click. No prompt.

3

Every number traces back to a source data row.

4

Root cause + confidence + what was ruled out.

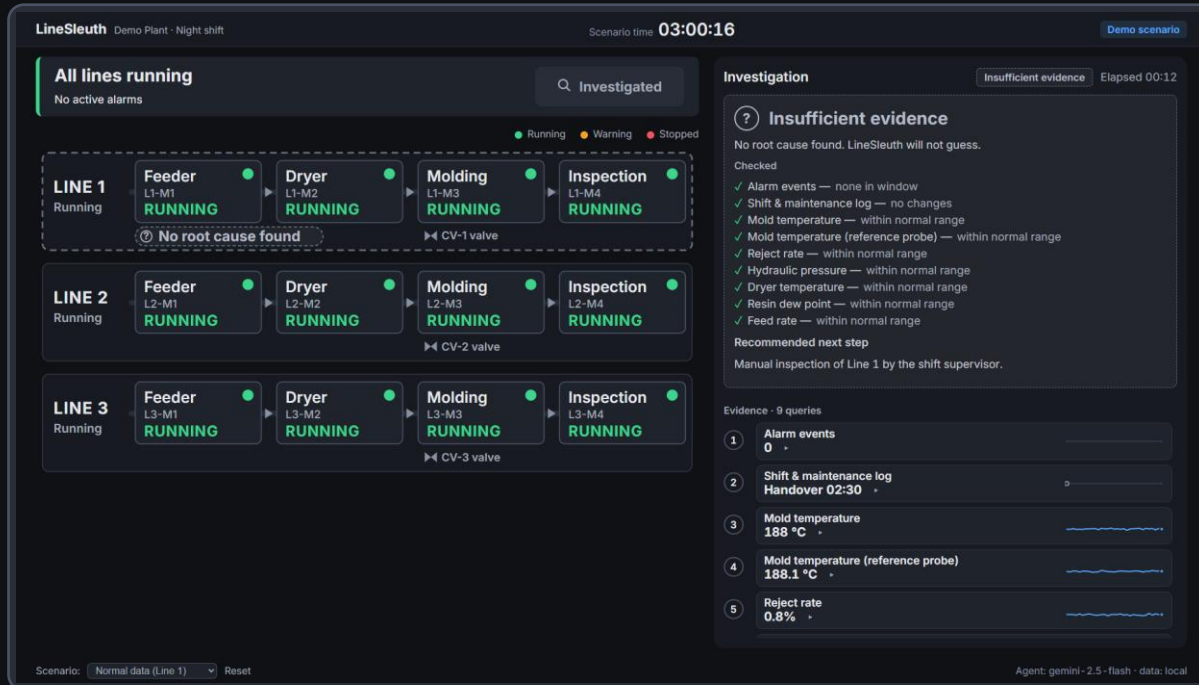
5

Before → after*

Backup video: stored on the laptop, not streamed.

* before-time pending interviews; after-time measured: median 12.6 s per investigation (gemini-2.5-flash, docs/qa/runs).

No evidence, no conclusion

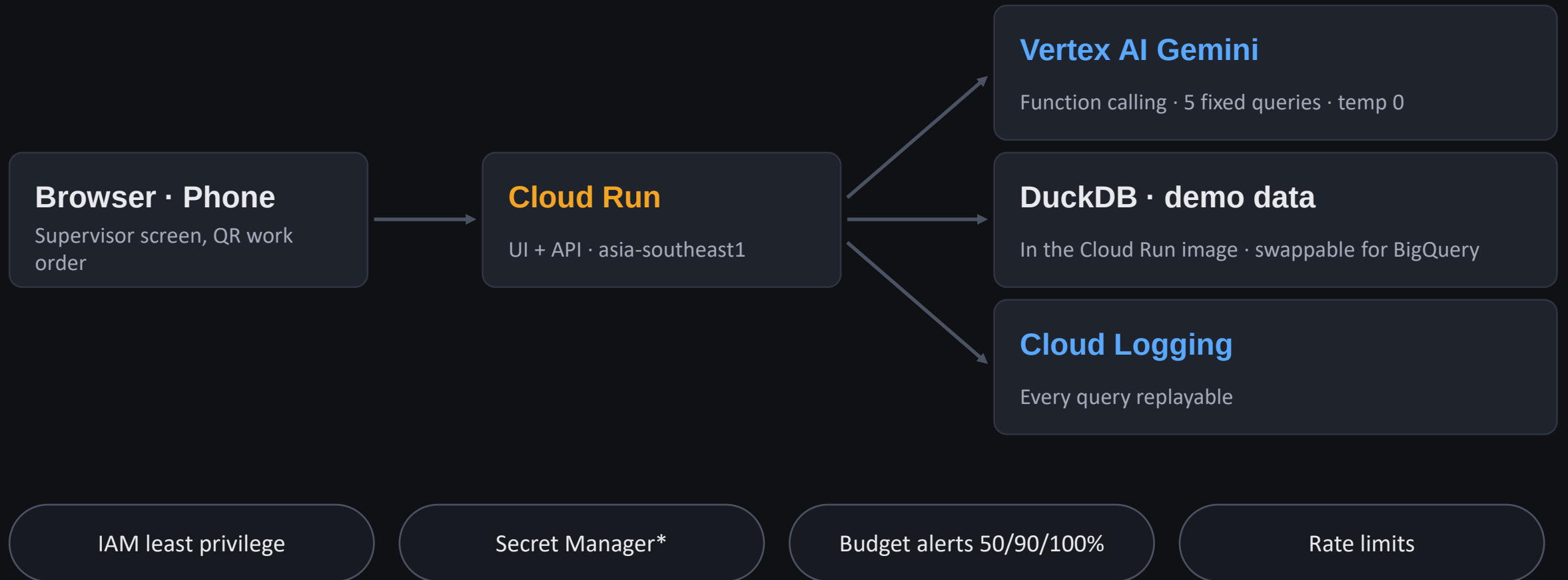


On healthy data LineSleuth answers “Insufficient evidence”, lists what it checked, and hands the decision back to a human.

- Gemini can only call fixed queries with checked parameters
- Numbers on cards come from source rows, not the model
- A root cause needs 2+ evidence cards
- Logs treated as untrusted data (prompt-injection tested)

ARCHITECTURE

Built on Google Cloud



** show Secret Manager only if the presenter key is stored there at submission. Model name only after it is verified.*

How we're different

	Siemens Industrial Copilot	Cognite	Google MDE	LineSleuth
Built for	Large plants in the Siemens ecosystem†	Asset-heavy enterprises with data teams†	Plants building a data platform on Google Cloud†	Small contract manufacturers without MES
Starts from	Automation / MES data†	Industrial data platform†	Connected factory data†	Excel sheets and PLC CSV exports
Job to be done	Broad industrial assistant	Industrial data operations	Data engine	Why did this line stop — with evidence
Pricing	Enterprise	Enterprise	Platform	Onboarding fee + per-line monthly

Google MDE is the platform our customers can grow into.

† competitor descriptions to verify against current public vendor material.

Business model

ONE-TIME

X*

Onboarding: turn Excel / PLC exports into clean data

PER LINE / MONTH

X*

Subscription, priced below the downtime and labour it saves

Value per line per month

incidents / month × time saved × downtime cost / hour × share of time spent searching

Roadmap

NOW · prototype

- One plant, simulated data
- Main story + grey-card test
- English work order via QR
- Regression set, 10 known root causes

NEXT

- Design-partner pilots on de-identified data
- Excel / PLC CSV import
- Vietnamese and Thai work orders
- Accounts and permissions

LATER

- Multiple lines and plants
- Multiple root causes per investigation
- Push notifications
- Early warning before a stoppage

Stoppage → root cause in ~12 s*

Every conclusion backed by evidence.

Hung Che Nick Lai

hongchelai@gmail.com · Live prototype: linesleuth-547147056278.asia-southeast1.run.app

** measured median 12.6 s per investigation (gemini-2.5-flash, 10 root causes × 3 runs, docs/qa/runs) · manual baseline pending interviews*

Validation status

☐ Regression set (10 known root causes, 3 distractors; bar $\geq 9/10$, 3 runs each) **10 / 10 (real Gemini)**

☐ Healthy data returns “Insufficient evidence” **2 / 2**

☐ Median time to conclusion (gemini-2.5-flash, docs/qa/runs) **12.6 s**

☐ Plant-manager interviews completed **[pending interviews]**

Known limitations

DATA

Simulated data; no live PLC / MES connection yet

SCOPE

One root cause per investigation

LANGUAGE

English only

SECURITY

No login in the prototype; not ready for real customer data